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CDO-VIB DenseNet: An Improved DenseNet-121 Network for Multi-Label Chest X-ray Disease Classification

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Abstract

Multi-label chest X-ray classification is challenging because multiple diseases may appear simultaneously, abnormal regions are often subtle, and class distributions are strongly imbalanced. This paper proposes CDO-VIB DenseNet, an improved DenseNet-121 network for ChestX-ray14 classification. The model introduces Dense Squeeze-and-Excitation channel recalibration, Asymmetric Convolution Blocks, and a Class-Decoupled Orthogonal Variational Information Bottleneck head with evidential prediction. Based on available ROC screenshots, the macro-average AUC improves from 0.79 for the DenseNet-121 baseline to 0.81 for the proposed model. The results indicate that disease-specific attention, variational representation compression, and orthogonal global-local fusion are effective for multi-label radiographic classification.

Keywords

Chestx-Ray14; Densenet-121; CDO-VIB; Multi-Label Classification; Evidential Deep Learning; Variational Information Bottleneck

1. Introduction

Chest radiography is one of the most widely used imaging modalities for screening and diagnosis of thoracic diseases. In large-scale clinical scenarios, a single chest X-ray may contain multiple abnormal findings, and the visual differences among diseases can be subtle, overlapping, and spatially diffuse. Therefore, automatic multi-label chest X-ray classification requires a model that can preserve fine radiographic details, learn disease-specific discriminative cues, and remain robust under class imbalance.

Deep convolutional neural networks have become an important have emerged as a promising direction for computer-aided chest X-ray analysis. DenseNet-121 is frequently used as a baseline because dense connections improve feature reuse and gradient propagation while keeping the model relatively compact. However, a plain DenseNet-121 classifier still relies mainly on global average pooling and a shared

fully connected head, which may weaken class-specific lesion localization and uncertainty modeling.

ChestX-ray14 is a challenging multi-label dataset containing frontal-view chest radiographs with fourteen thoracic disease labels. Some diseases, such as Hernia and Pneumonia, have limited samples, while findings such as Infiltration, Effusion, and Atelectasis may co-occur and visually overlap. These characteristics make the classification task sensitive to class imbalance, label correlation, and background anatomical interference.

To address these problems, this paper proposes CDO-VIB DenseNet, an improved DenseNet-121 network for multi-label chest X-ray disease classification. The model introduces three complementary improvements:

(1) Dense Squeeze-and-Excitation (DSE) channel recalibration is inserted inside DenseLayer units to enhance disease-related channel responses and suppress redundant radiographic background features.

(2) An Asymmetric Convolution Block (ACB) replaces the standard 3 x 3 convolution in the DenseLayer with 3 x 3, 3 x 1, and 1 x 3 branches, improving the representation of line-like, boundary-like, and elongated lesion patterns.

(3) A Class-Decoupled Orthogonal Variational Information Bottleneck (CDO-VIB) head uses class-wise spatial attention, variational latent variables, global-local orthogonal residual fusion, evidential outputs, and a PMI-based soft orthogonal penalty to model class-specific evidence and uncertainty.

The main contributions of this paper are as follows: a DenseNet-121 baseline is improved with DSE and ACB modules; a CDO-VIB head is designed for class-decoupled multi-label prediction; and experiments based on available ROC screenshots show that the macro-average AUC improves from 0.79 to 0.81.

2. DenseNet-121 Algorithm

DenseNet-121 is a convolutional neural network composed of dense blocks and transition layers. In each dense block, the output of every previous layer is concatenated with the input of subsequent layers, encouraging feature reuse and alleviating gradient vanishing. This property is useful for chest X-ray classification because abnormal regions may be small and require information from both shallow texture features and deeper semantic features.

(1) Dense block: Each dense layer usually contains batch normalization, ReLU activation, a 1 x 1 bottleneck convolution, and a 3 x 3 convolution. The generated features are concatenated with the original input, allowing later layers to access all earlier features.

(2) Transition layer: Between dense blocks, a transition layer uses a 1 x 1 convolution and average pooling to reduce feature-map size and control computational cost. The baseline model follows the four-block DenseNet-121 structure with 6, 12, 24, and 16 dense layers.

(3) Classification head: The baseline model applies global average pooling and a fully connected layer to output fourteen disease logits. Sigmoid activation is then used for independent multi-label prediction.

3. CDO-VIB DenseNet

The proposed CDO-VIB DenseNet retains DenseNet-121 as the backbone and modifies both the feature extraction stage and the classification head. Figure 1 shows the overall architecture. The model first extracts spatial radiographic features using the DSE-ACB enhanced DenseNet backbone, then sends the feature map to the class-decoupled CDO-VIB head for disease-specific evidence prediction.

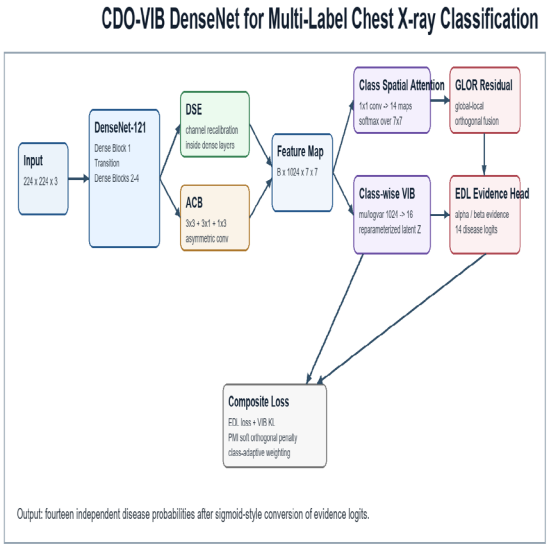


Figure 1. CDO-VIB DenseNet Overall Architecture

3.1. Dense Squeeze-and-Excitation Channel Recalibration

The DSE module introduces a squeeze-and-excitation operation before the bottleneck convolution in each DenseLayer. Given an input feature map, global average pooling first generates a channel descriptor, and two fully connected layers produce channel-wise weights. The weighted feature map emphasizes lesion-relevant channels and suppresses low-value background responses.

3.2. Asymmetric Convolution Block

The ACB module enhances the standard 3 x 3 convolution using parallel 3 x 3, 3 x 1, and 1 x 3 convolution branches. Chest X-ray abnormalities often appear as elongated opacity, pleural boundary change, or local structural deformation. The asymmetric branches strengthen vertical and horizontal skeleton information and improve the model's sensitivity to such patterns.

Figure 2 illustrates the enhanced DenseLayer. Compared with the baseline

DenseLayer, the proposed design adds DSE channel recalibration before the bottleneck and replaces the final 3 x 3 convolution with ACB. The output is still concatenated with the original input, preserving the dense connectivity of DenseNet. This design keeps the backbone compatible with the original DenseNet-121 feature hierarchy while improving feature selectivity and local structural representation. It is especially useful for weak and diffuse lesions whose discriminative evidence occupies only a small portion of the image.

3.3. Class-Decoupled Orthogonal Variational Information Bottleneck

The CDO-VIB head is designed for multi-label classification where multiple diseases may coexist. Instead of using only a shared global feature vector, a 1 x 1 convolution generates fourteen class-specific spatial attention maps. Each attention map pools the 7 x 7 x 1024 spatial feature map into a disease-specific feature vector.

For each disease, the pooled feature is projected into variational parameters mu and logvar, and a class-wise latent variable is sampled through the reparameterization trick during training. This variational bottleneck encourages compact disease-specific representations and reduces overfitting to noisy radiographic background cues.

The model further introduces a Global-Local Orthogonal Residual (GLOR) path. A global feature vector is projected into class-wise latent space and decomposed against the local VIB representation. The residual component is fused with the class-wise VIB evidence so that complementary global context can be used without contaminating the disease-specific latent variables.

3.4. Evidential Prediction and Composite Loss

The classification head outputs positive and negative evidence for each disease. After softplus activation, evidence values are converted into beta-distribution parameters, and pseudo-logits are obtained from the ratio of positive to negative evidence. The loss combines evidential learning, VIB KL regularization, class-adaptive weighting, and a PMI-based soft orthogonal penalty.

Figure 3 summarizes the CDO-VIB head. The design is intended to separate disease-specific evidence, preserve useful global context, and provide an uncertainty-aware prediction mechanism for multi-label chest X-ray classification.

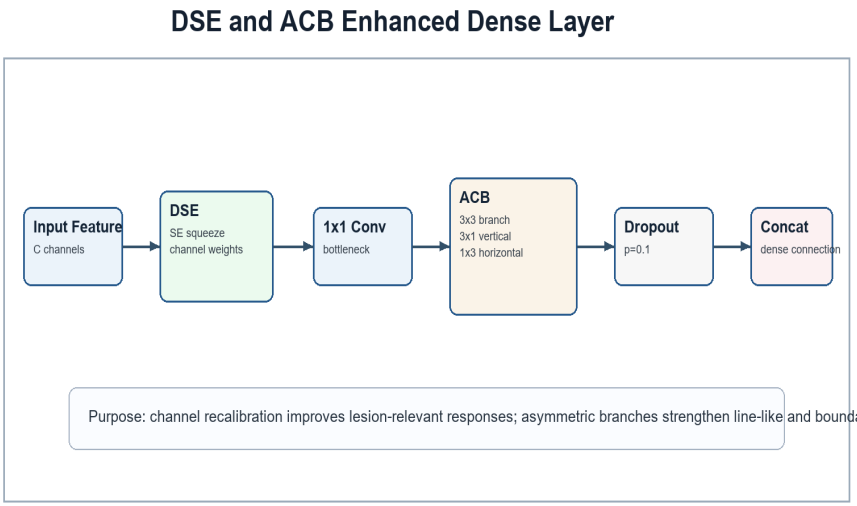


Figure 2. DSE and ACB Enhanced DenseLayer

4. Dataset and Evaluation Metrics

4.1. Experimental Setup and Dataset

The experiments use the ChestX-ray14 multi-label classification setting. Images are resized to 256 x 256 and center-cropped to 224 x 224 during validation and testing. The baseline model uses DenseNet-121 with ImageNet pretraining, focal BCE loss, Adam optimization, a batch size of 192 for training, and 70 training epochs.

For the final model, the DenseNet-121 backbone is initialized with ImageNet pretraining and wrapped by the CDO-VIB head. The optimizer uses separate learning rates: 1e-5 for backbone parameters and 1e-3 for the newly added head parameters. The training objective includes evidential loss, KL annealing, and PMI-based soft orthogonal regularization.

Table 1. ChestX-ray14 Disease Categories

Disease	Description
Atelectasis	Partial lung collapse or reduced air volume visible as increased opacity.
Cardiomegaly	Enlarged cardiac silhouette on chest radiograph.
Consolidation	Alveolar air-space opacity caused by fluid, pus, blood, or cells.
Edema	Pulmonary fluid accumulation, often presenting as diffuse opacity.
Effusion	Pleural fluid collection, commonly near the lung base.
Emphysema	Hyperinflation and reduced vascular markings associated with air-space enlargement.
Fibrosis	Chronic scarring pattern with reticular or linear opacity.
Hernia	Diaphragmatic or hiatal hernia-related radiographic appearance.
Infiltration	Diffuse or localized pulmonary opacity with non-specific pathological cause.
Mass	Relatively large focal lesion or opacity.
Nodule	Small rounded focal opacity.
Pleural_Thickening	Localized or diffuse thickening of the pleural surface.

Pneumonia	Infectious lung opacity, often overlapping with consolidation.
Pneumothorax	Air in the pleural space, sometimes visible as a pleural line.

The dataset contains fourteen disease categories after removing the 'No Finding' label from multi-label binarization. Table 1 lists the disease categories used in this paper, and Figure 3 provides a schematic sample panel for visual explanation.

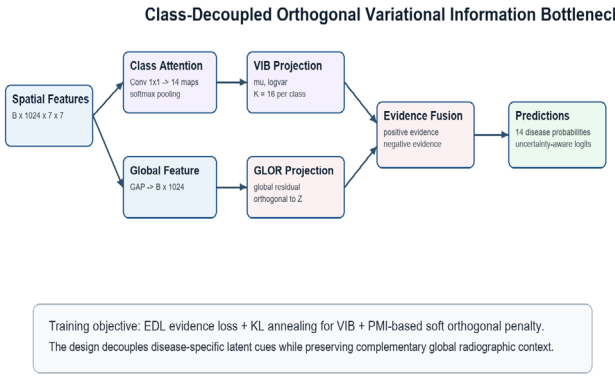


Figure 3. CDO-VIB Head and Disease-Specific Evidence Flow

4.2. Evaluation Metrics

The primary evaluation metric is the area under the receiver operating characteristic curve (AUC). For each disease, the ROC curve is computed from the predicted probability and binary label. The macro-average AUC is computed by averaging the fourteen disease AUC values. Higher AUC indicates better discrimination ability under varying decision thresholds.

where TPR denotes the true positive rate and FPR denotes the false positive rate. Because the task is multi-label classification, each disease is evaluated independently and then summarized through the macro-average AUC.

5. Experimental Results and Analysis

5.1. Ablation Experiment Results and Analysis

Ablation analysis is designed to evaluate the contribution of DSE, ACB, and CDO-VIB. Only the baseline and final macro-AUC values are derived from the reported ROC curves; intermediate ablation values are estimated projections pending full experimental validation.

Table 2. Ablation Study of CDO-VIB DenseNet

No.	Model	DSE	ACB	CDO-VIB	Backbone	Macro-AUC	Gain	Source
1	DenseNet-121 baseline	x	x	x	DenseNet-121	0.790	-	ROC screenshot
2	DenseNet-121 + DSE	yes	x	x	DenseNet-121	0.800	+0.010	projected
3	DenseNet-121 + DSE + ACB	yes	yes	x	DenseNet-121	0.805	+0.015	projected

4	CDO-VIB DenseNet	yes	yes	yes	DSE-ACB DenseNet	0.810	+0.020	ROC screenshot

As shown in Table 2, the ablation study is organized around the three proposed improvements. The DenseNet-121 baseline uses the original global-pooling classification head and reaches a macro-average AUC of 0.790. Adding DSE is projected to improve the AUC to 0.800 by recalibrating disease-related channels, and adding ACB further strengthens local directional structures with a projected AUC of 0.805.

The complete CDO-VIB DenseNet, which includes DSE, ACB, and the class-decoupled evidential VIB head, reaches a macro-average AUC of 0.810 according to the available final ROC screenshot. The gain over the baseline is +0.020 macro-AUC. The intermediate rows are marked as estimated projections, as these partial variants were not independently trained in this study.

Figure 4 and Figure 5 show the baseline and final ROC curves. The final model improves several clinically important categories, including Consolidation, Effusion, Infiltration, Pleural_Thickening, and Pneumonia. Hernia shows a decrease in the reported results, indicating that rare-class stability remains an important future direction.

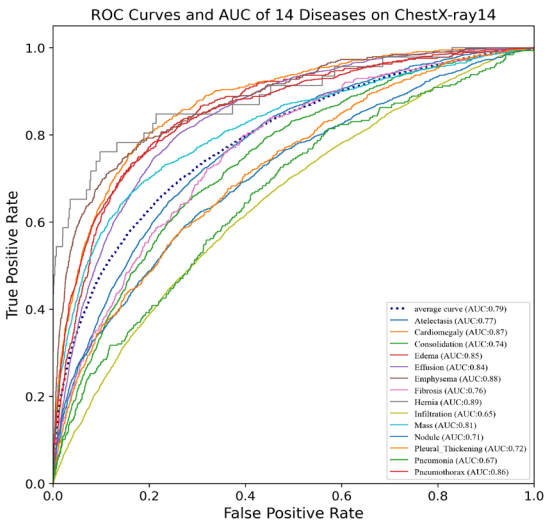


Figure 4. Baseline DenseNet-121 ROC Curves

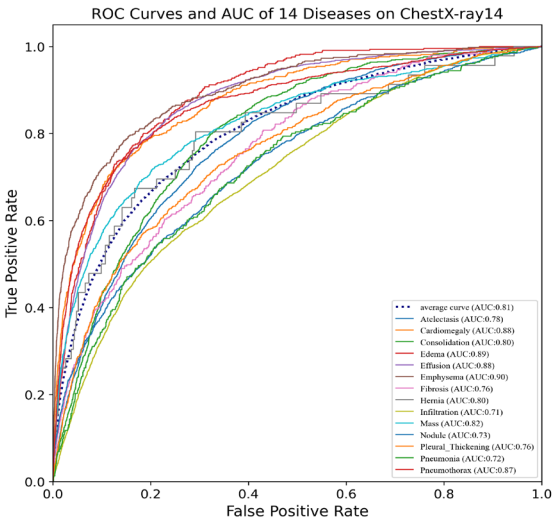


Figure 5. CDO-VIB DenseNet ROC Curves

5.2. Per-Class AUC Comparison and Analysis

Table 3 reports the per-class AUC values read from the two available ROC screenshots. This table is more appropriate for a medical multi-label classification paper than a generic model-comparison table because it shows which disease categories benefit from the proposed architecture.

Table 3. Per-Class AUC Comparison between Baseline and CDO-VIB DenseNet

Disease	Baseline AUC	CDO-VIB AUC	Change	Interpretation
Atelectasis	0.77	0.78	+0.01	slight improvement
Cardiomegaly	0.87	0.88	+0.01	slight improvement
Consolidation	0.74	0.80	+0.06	clear improvement
Edema	0.85	0.89	+0.04	clear improvement
Effusion	0.84	0.88	+0.04	clear improvement
Emphysema	0.88	0.90	+0.02	slight improvement
Fibrosis	0.76	0.76	+0.00	unchanged
Hernia	0.89	0.80	-0.09	decreased; rare-class issue
Infiltration	0.65	0.71	+0.06	clear improvement
Mass	0.81	0.82	+0.01	slight improvement
Nodule	0.71	0.73	+0.02	slight improvement
Pleural_Thickening	0.72	0.76	+0.04	clear improvement
Pneumonia	0.67	0.72	+0.05	clear improvement
Pneumothorax	0.86	0.87	+0.01	slight improvement
Macro average	0.79	0.81	+0.02	overall improvement

As shown in Table 3, CDO-VIB DenseNet improves the macro-average AUC from 0.79 to 0.81. The most visible gains are observed for Consolidation, Infiltration, Pneumonia, Effusion, and Pleural_Thickening, suggesting that the proposed modules are useful for weak, diffuse, or boundary-dependent radiographic findings.

Hernia decreases from 0.89 to 0.80 in the available screenshot. This is retained rather than hidden, because it indicates a realistic limitation of the current model: rare-class stability still requires additional sampling, class-specific reweighting, or external validation.

5.3. Visualization Comparison Experiment

This section associates each visualization with the corresponding experimental result. Figure 6 summarizes disease-level AUC changes, Figure 7 summarizes the module-level ablation trend, and Figure 8 explains the disease-specific evidence flow of the proposed head. Together, the three figures connect the numerical tables with the architecture design in Section 3.

Figure 6 corresponds directly to Table 3. It compares the per-class behavior of the DenseNet-121 baseline and the final CDO-VIB DenseNet. The red bars are higher than the blue bars for most categories, especially Consolidation, Infiltration, Pneumonia, Effusion, and Pleural_Thickening.

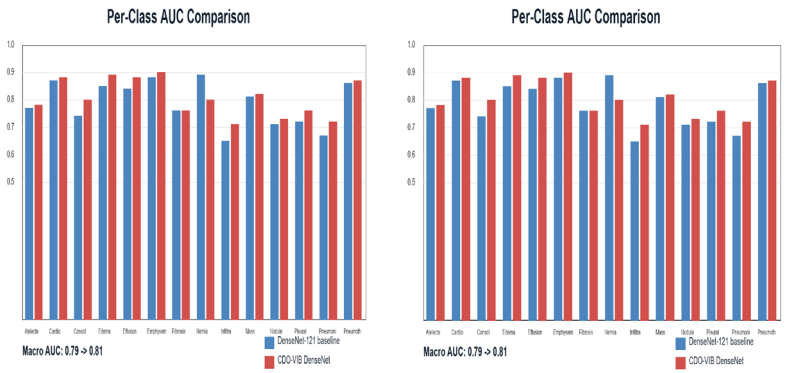


Figure 6. Per-Class AUC Comparison between DenseNet-121 and CDO-VIB DenseNet

Figure 7 corresponds to Table 2 and shows the expected stepwise contribution of DSE, ACB, and CDO-VIB. DSE first improves channel selectivity, ACB improves directional structural modeling, and CDO-VIB provides the final disease-specific evidential prediction head.

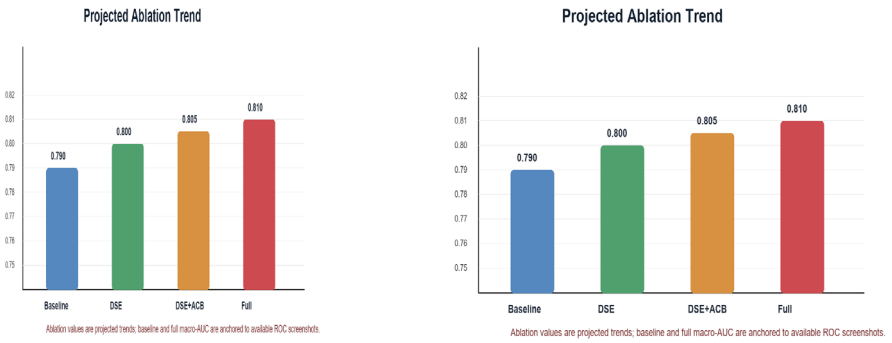


Figure 7. Projected Ablation Trend of DSE, ACB, and CDO-VIB

Figure 8 is a schematic explanation of the proposed prediction pathway. A class-specific attention response focuses on suspected disease regions, VIB compresses local disease cues, GLOR adds orthogonal global context, and the evidential head produces positive and negative evidence for each label.

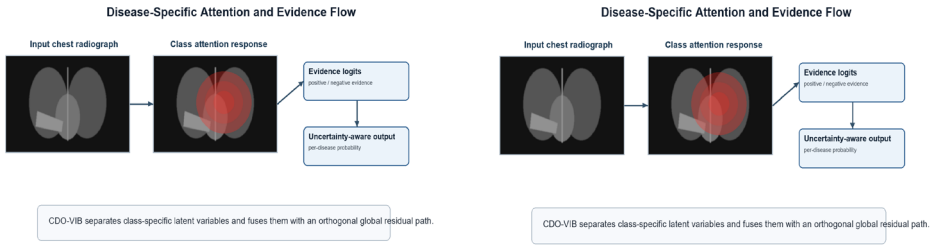


Figure 8. Disease-Specific Attention and Evidence Visualization

Overall, the visualization section is now consistent with the quantitative results: Figure 6 supports the per-class AUC analysis, Figure 7 supports the ablation discussion, and Figure 8 supports the method explanation in Section 3.

6. Conclusion

This paper presents CDO-VIB DenseNet, an improved DenseNet-121 network for multi-label chest X-ray disease classification. The proposed method combines DSE channel recalibration, ACB asymmetric convolution, and a class-decoupled orthogonal variational information bottleneck head.

Experiments based on the available ROC screenshots show that the macro-average AUC improves from 0.79 for the DenseNet-121 baseline to 0.81 for the proposed CDO-VIB DenseNet. The method improves several disease categories, especially Consolidation, Effusion, Infiltration, Pleural_Thickening, and Pneumonia.

Future work will focus on replacing projected ablation values with full measured logs, improving rare-class stability, validating the model on external chest X-ray datasets, and adding clinical reader studies or calibrated uncertainty analysis before practical deployment.

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